Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 8.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Given,

$$(a^6)^{-\frac{2}{3}} = \frac{1}{a^m}$$

Assuming $a \neq 0$ and $a \neq 1$, what is the value of m?

A 2 **B** 3 **C** 4 **D** 5 **E** 6

2. The sum of three numbers is 28. The second number is three times the first and the third is seven less than the second .

What are the numbers?

A $5, 10, 8$ **B** $5, 12, 8$ **C** $5, 15, 10$ **D** $5, 15, 8$ **E** $4, 15, 8$

3. Stella has 5 times as many books as Tina. Hypothetically, if Stella were to give 16 books to Tina, they would each have the same number.

How many books does Stella have?

A 25 **B** 30 **C** 35 **D** 40 **E** 45

4. A tortoise makes a journey in two parts; it can either walk at $4\ m/s$ or crawl at $3\ m/s$. If the tortoise walks the first part and crawls the second, it takes $110\ s$. If it crawls the first part and walks the second, it takes $100\ s$.

Find the lengths of the two parts of the journey.

A $80m, 100m$ **B** $100m, 120m$ **C** $120m, 240m$ **D** $240m, 480m$ **E** $40m, 80m$

5. In three years time a pet cat will be as old as his owner was four years ago. Their present ages total 13 years.

Find the age of the cat now.

A 1 year **B** 2 years **C** 3 years **D** 4 years **E** 5 years

6. A box contains balls of different colours. The box is opened and a ball is selected at random. The probability that the ball is white is 0.9 and the probability that the ball is black is 0.04

Find the probability that the ball is neither white nor black.

A 0.03 **B** 0.06 **C** 0.12 **D** 0.24 **E** 0.48

7. It is seen that $\frac{4\sqrt{3}+5\sqrt{60}}{2\sqrt{3}}$ can be written in the form $a + b\sqrt{b}$.

What are the values of a and b?

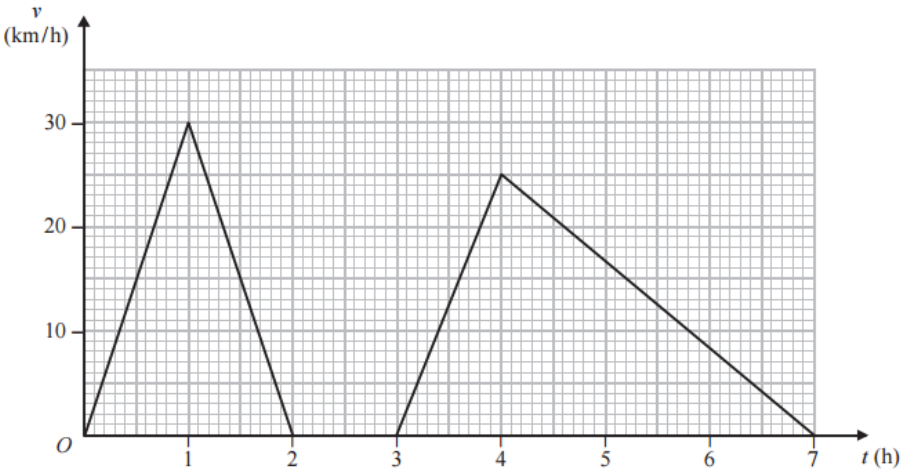
A $a = 1, b = 2$ **B** $a = 5, b = 2$ **C** $a = 2, b = 5$ **D** $a = 2, b = 1$ **E** $a = 3, b = 5$

8. An increase of speed of $4\ km/h$ on a journey of $32\ km$ reduces the time taken by $4\ hours$.

Find the original speed.

A $2\ km/h$ **B** $4\ km/h$ **C** $6\ km/h$ **D** $8\ km/h$ **E** $10\ km/h$

9. The speed-time graph shows information about the motion of a particle in a $7\ hour$ period.



Calculate the total distance, in km, travelled by the particle in the $7\ hours$.

A $20\ km$ **B** $40\ km$ **C** $80\ km$ **D** $160\ km$ **E** $320\ km$

10. A simultaneous equation is shown

$$4x - 0.5y = 12.5$$

$$3x + 0.8y = 8.2$$

Find the values of x and y.

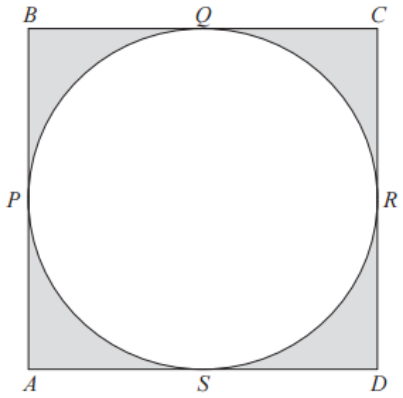
A $x = -1, y = 3$ **B** $x = 1, y = 3$ **C** $x = -1, y = -3$ **D** $x = 3, y = -1$ **E** $x = 3, y = 1$

11. The height of cone A is $6\ cm$ and the height of a similar cone B is $10\ cm$. The volume of cone A is $189\ cm^3$

Calculate the volume, in cm^3, of cone B.

A $437.5\ cm^3$ **B** $875\ cm^3$ **C** $1750\ cm^3$ **D** $52.5\ cm^3$ **E** $625\ cm^3$

12. $ABCD$ is a square of side $12\ cm$. The circle $PQRS$ touches the sides of the square at P, Q, R and S as shown in the diagram.



Find, in terms of π, the total area, in cm^2, of the shaded region in the diagram.

A $(144 - 36\pi^2)$ cm^2 **B** $36(4 - 3\pi)$ cm^2 **C** $36(4 - 2\pi)$ cm^2 **D** $36(4 - \pi)$ cm^2 **E** $36(8 - \pi)$ cm^2

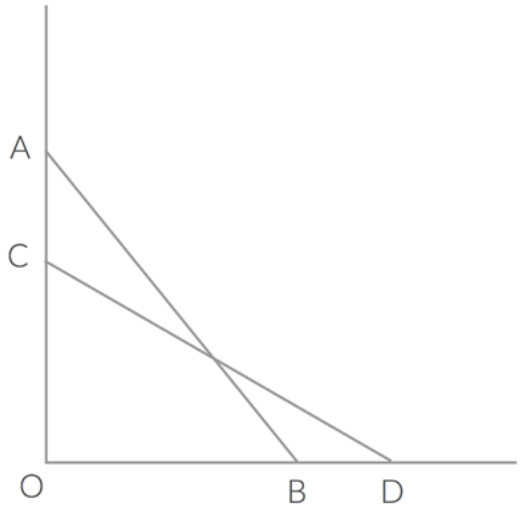
13. Write down the range of values of x which satisfies both the inequalities

$$3x - 7 < 3 - x$$

$$x - 12 \leq 3(x - 3)$$

A -1.5 **B** 1.5 **C** -2.5 **D** $-1.5 \leq x < 2.5$ **E** -1.5

14. The diagram represents the starting position (AB) and the finishing position (CD) of a ladder as it slips. The ladder is leaning against a vertical wall.



Given, $AC = x$, $OC = 4AC$, $BD = 2AC$ and $OB = 5m$.

Find the length of the ladder.

A $10.3\ m$ **B** $20.6\ m$ **C** $41.2\ m$ **D** $82.4\ m$ **E** $164.8\ m$

15. Make y the subject in the equation.

$$\frac{m(ny - e^2)}{p} + n = 5n$$

A $\frac{4n+me^2}{mn}$ **B** $4\left(\frac{np+me^2}{mn}\right)$ **C** $\frac{4np+2me^2}{mn}$ **D** $\frac{4np+me^2}{mn}$ **E** $\frac{2np+me^2}{mn}$

16. 1050 Chinese yuan (¥) is to be divided between three people in the ratios $3 : 5 : 7$

Find the difference, in ¥, between the largest share and the smallest share.

A 140 ¥ **B** 280 ¥ **C** 420 ¥ **D** 560 ¥ **E** 270 ¥

17. The height of a right circular cylinder is four times the radius of the base of the cylinder. The volume of the cylinder is $500\pi\ cm^3$.

Calculate, in cm, the radius of the base of the cylinder.

A $25\ cm$ **B** $5\ m$ **C** $5\ cm$ **D** $25\ m$ **E** $125\ cm$

18. Evaluate, with $x = 16$ and $y = 8$,

$$\frac{(x^{0.25} + y^{-1})}{x^{0.25}}$$

A $19/16$ **B** $5/16$ **C** 1 **D** $17/16$ **E** $9/16$

19. The point A has coordinates $(-3, 7)$ and the point B has coordinates $(5, -8)$

What is the gradient of AB?

A $-15/16$ **B** $-30/8$ **C** $-15/32$ **D** $-15/8$ **E** $15/8$

20. Given, $3y - 4 < 12 < 4y - 5$

Find an integer value of y.

A 4 **B** 5 **C** -5 **D** -4 **E** 3

21. Expand and simplify

$$(x - y)(y + z) - (x + y)(y - z)$$

A $2xz$ **B** $2xz + 2y^2$ **C** $2xz - 2y^2$ **D** $2y^2 - 2xz$ **E** $xz - y^2$

22. A bag contains 10 counters. Of these counters, 7 are black and 3 are yellow. Two of these counters are to be taken at random, without replacement, from the bag.

Find the probability that the two counters taken are of different colours.

A $7/30$ **B** $7/15$ **C** $14/15$ **D** $7/60$ **E** $14/45$

23. Find the largest prime number x such that $x^2 \leq 400$

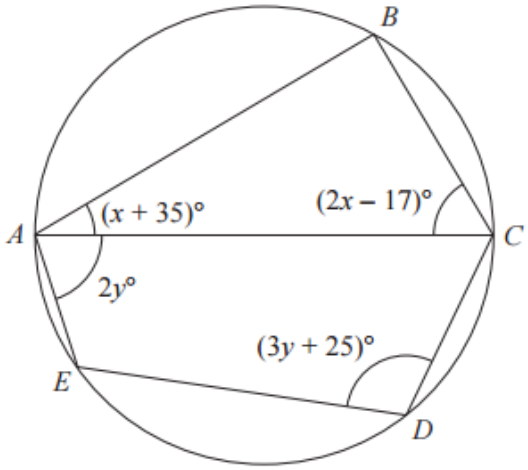
A 1 **B** 7 **C** 13 **D** 17 **E** 19

24. A straight line joins the points $(2m, n)$ and $(3, -4)$.

What is the value of n if the line is parallel to the x-axis?

A 2 **B** -2 **C** 4 **D** -4 **E** 3

25. $ABCDE$ is a circle with diameter AC. $\angle BAC = (x + 35)°$ and $\angle BCA = (2x - 17)°$



What is the value of x?

A 6 **B** 12 **C** 24 **D** 48 **E** 36

26. A high-speed train runs between the cities of London and Chard. The distance between these two cities is $400\ km$. A train leaves London for Chard and travels at an average speed of $x\ km/h$. On the return journey from Chard to London, the train travels at an average speed of $(x - 60)\ km/h$. The train took 20 minutes longer to return to London than it took to travel from London.

From the given information, which one is the correct equation about the journey times?

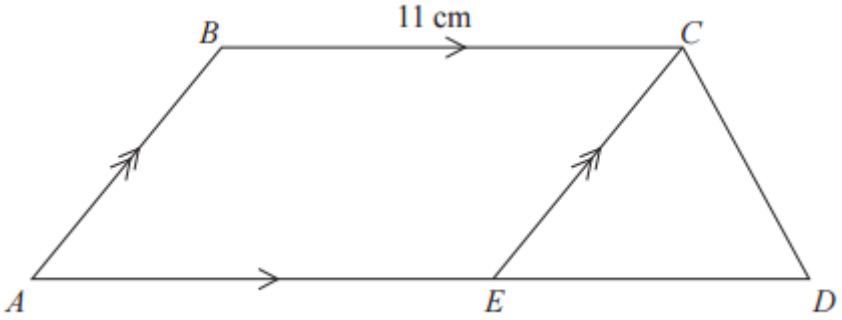
A $\frac{400}{x-60} - \frac{400}{x} = \frac{1}{3}$ **B** $\frac{400}{x-60} - \frac{400}{x} = \frac{1}{6}$ **C** $\frac{200}{x-60} - \frac{400}{x} = \frac{1}{3}$ **D** $\frac{400}{x} - \frac{400}{x-60} = \frac{1}{3}$ **E** $\frac{400}{x+60} - \frac{400}{x} = \frac{1}{3}$

27. Given that $x = 2 \times 10^{10}$ and $y = 16 \times 10^{11}$

Calculate the value of $\frac{x}{y}$. Give your answer in standard form.

A $1.25\ x\ 10^{-3}$ **B** $1.50\ x\ 10^{-2}$ **C** $1.25\ x\ 10^{-2}$ **D** $2.50\ x\ 10^{-2}$ **E** $2.50\ x\ 10^{-3}$

28. $ABCD$ is a trapezium with $BC = 11\ cm$, $AD = 15\ cm$ and BC parallel to AD. The point E on AD is such that EC is parallel to AB. The area of triangle $ECD = 7\ cm^2$.



Find the area, in m^2 , of the trapezium $ABCD$.

A $11.375\ cm^2$ **B** $22.75\ cm^2$ **C** $45.5\ cm^2$ **D** $91\ cm^2$ **E** $182\ cm^2$

29. In the figure, O is the centre of the circle BCD. The tangent to the circle at D is ADE and $ABOC$ is a straight line.

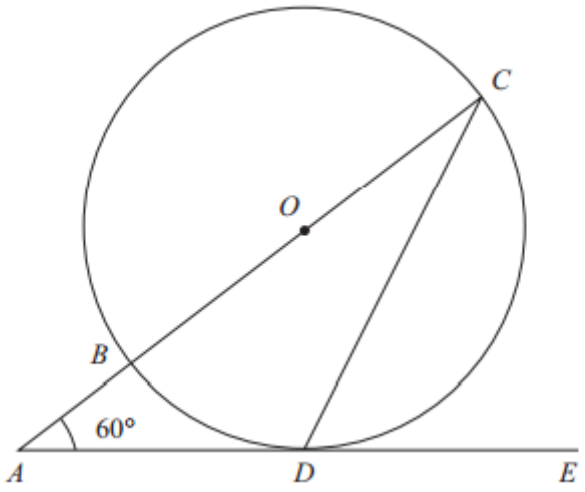


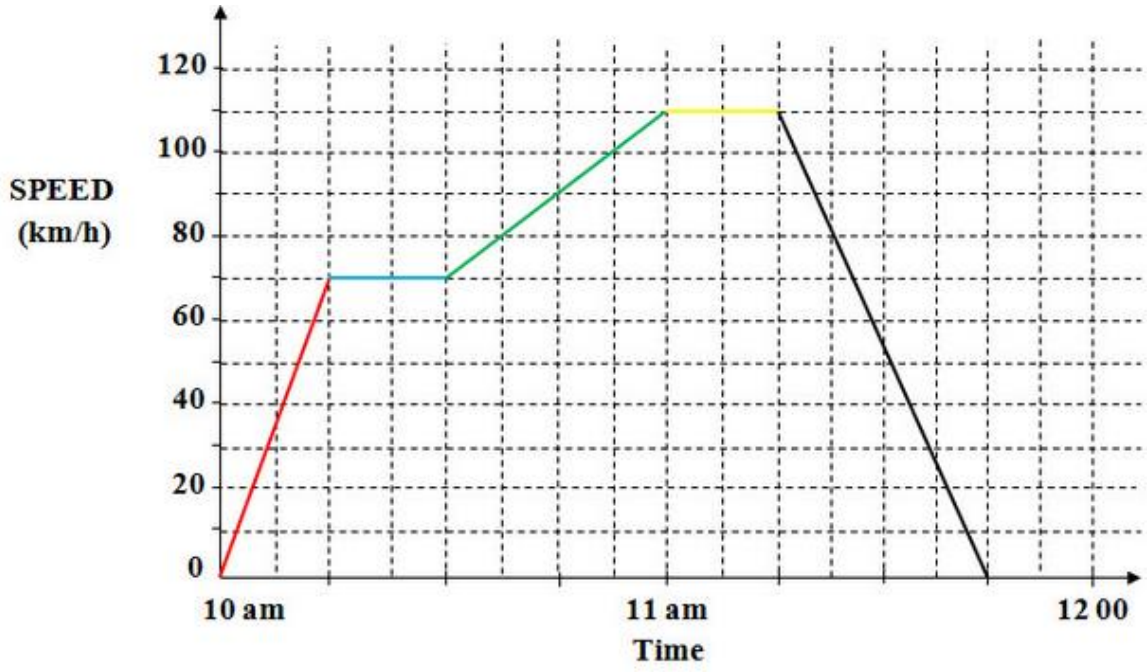
Diagram **NOT** accurately drawn

Given that $\angle CAD = 60°$,

calculate the size, in degrees, of $\angle CDE$.

A 37.5° **B** 75° **C** 150° **D** 60° **E** 80°

30. The diagram shows the speed-time graph of a train moving from Birmingham to London.



What is the total distance travelled between $10\ am$ and $11\ am$?

A $71.25\ km$ **B** $61.25\ km$ **C** $36.25\ km$ **D** $126.25\ km$ **E** $98.75\ km$

Answer key:

№	Correct Answer	Your answer
1	C	
2	D	
3	D	
4	C	
5	C	
6	B	
7	C	
8	B	
9	C	
10	D	
11	B	
12	D	
13	D	
14	B	
15	D	
16	B	
17	C	
18	D	
19	D	
20	B	
21	C	
22	B	
23	E	
24	D	
25	C	
26	A	
27	C	
28	C	
29	B	
30	A	
		Total: _____